

The empirical recurrence for OEIS A304672, recovered from its tail

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Abstract

OEIS A304672 records “Empirical recurrence of order 70” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{70} = 7168 \neq 0$, so no further tail satisfies anything shorter; any order-70 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304672 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 0, 2, 3, 5 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 13, 23, 56, 199, 598, 1950, 6588, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (May 16 2018 14:44:53, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 70: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 70.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 70 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	7	c_{19}	−78200	c_{37}	407059	c_{55}	13540198
c_2	−14	c_{20}	−19392	c_{38}	−7673798	c_{56}	25576038
c_3	11	c_{21}	150207	c_{39}	29679393	c_{57}	−32878774
c_4	−36	c_{22}	−212991	c_{40}	−5326227	c_{58}	14675586
c_5	52	c_{23}	93498	c_{41}	−48020594	c_{59}	−10339840
c_6	48	c_{24}	−386536	c_{42}	8072675	c_{60}	4482175
c_7	13	c_{25}	760494	c_{43}	68468895	c_{61}	2760110
c_8	−142	c_{26}	157367	c_{44}	17264443	c_{62}	−2314140
c_9	136	c_{27}	−890213	c_{45}	−54487947	c_{63}	−258072
c_{10}	−150	c_{28}	224242	c_{46}	13121874	c_{64}	452664
c_{11}	116	c_{29}	−1141426	c_{47}	−8347159	c_{65}	109440
c_{12}	−1340	c_{30}	1655984	c_{48}	−41662457	c_{66}	−267824
c_{13}	−2094	c_{31}	−1288931	c_{49}	18774753	c_{67}	139808
c_{14}	4531	c_{32}	4000312	c_{50}	20822899	c_{68}	4096
c_{15}	−8296	c_{33}	3290063	c_{51}	76047956	c_{69}	−28928
c_{16}	29855	c_{34}	2083092	c_{52}	−100415029	c_{70}	7168
c_{17}	−10003	c_{35}	−7125913	c_{53}	3365589		
c_{18}	25467	c_{36}	−11099257	c_{54}	4425122		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A304672 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{70} - c_1 x^{70-1} - \dots - c_{70}$. Its constant term is $-c_{70} = -7168$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 70 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 70, so $\deg q = 70$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 70, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 70, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 7168.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-70 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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